THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Mendel chose the garden pea for his experiments because 1

(a) it has many contrasting characters and is easy to grow
(b) it grows only in cold regions
(c) it reproduces only asexually
(d) it has no flowers

2. In muscles during vigorous exercise, glucose breaks down anaerobically to give: 1

(a) lactic acid **(b)** CO_2 only **(c)** water **(d)** ethanol

3. The roots of a plant grow downward in response to: 1

(a) gravity **(b)** water only **(c)** air **(d)** light

4. The part of the flower that becomes the fruit is the 1

(a) ovule **(b)** style **(c)** stigma **(d)** ovary

5. The F2 phenotypic ratio in the dihybrid cross is 1

(a) 1 : 1 : 1 : 1 **(b)** 9 : 3 : 3 : 1 **(c)** 1 : 2 : 1 **(d)** 3 : 1

6. Urine is stored temporarily in: 1

(a) urinary bladder **(b)** ureter **(c)** kidney **(d)** urethra

7. The master gland of the body is the: 1

(a) thyroid **(b)** pituitary **(c)** pancreas **(d)** adrenal

8. The testes lie outside the abdominal cavity in the scrotum because 1

(a) they are too large for the abdomen
(b) they need more oxygen
(c) they produce urine
(d) sperm formation needs a temperature lower than body temperature

9. A normal human female has the sex chromosomes 1

(a) XX **(b)** XY **(c)** YY **(d)** XO

10. Which of the following are required for autotrophic nutrition in green plants? 1

(a) only carbon dioxide and water
(b) carbon dioxide, water, chlorophyll and sunlight
(c) only chlorophyll and sunlight
(d) only sunlight

11. Which of the following is not a plant hormone? 1

(a) auxin **(b)** cytokinin **(c)** gibberellin **(d)** insulin

12. Hydra reproduces asexually mainly by 1

(a) spore formation **(b)** binary fission **(c)** multiple fission **(d)** budding

13. The transmission of characters from parents to offspring is called 1

(a) heredity **(b)** regeneration **(c)** evolution **(d)** variation

14. In yeast, anaerobic respiration produces: 1

(a) glucose **(b)** lactic acid
(c) water **(d)** ethanol and carbon dioxide

15. Tendrils coil around a support due to: 1

(a) chemotropism **(b)** thigmotropism **(c)** geotropism **(d)** phototropism

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the role of the brain in reflex action? 2

17. What is vegetative propagation? Give one example. 2

18. What is a monohybrid cross? Give the F2 ratio. 2

19. What are the components of the transport system in highly organised plants? 2

20. Why is the use of iodised salt advisable? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why the father, not the mother, determines the sex of the child. 3

22. Compare aerobic and anaerobic respiration with one product each. 3

23. State the functions of the cerebrum, cerebellum and medulla. 3

24. What changes occur in boys and girls at puberty? Give two for each. 3

OR

What is the role of DNA in heredity? 3

25. How do guard cells regulate the opening and closing of stomatal pores? 3

OR

Why is the reflex arc important? Give an example. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe excretion in human beings: parts of the system and the working of the nephron. 5

OR

Explain how auxin brings about phototropic bending of a shoot. 5

27. Describe the human female reproductive system. Where does fertilisation occur and where does the zygote develop? 5

OR

Explain Mendel's experiments and the conclusions drawn from them. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND